

Project Sprint #1

The Solitaire game is described in CS449HomeworkOverview. You should read the description very carefully.

In this assignment, you aim to specify the requirements (i.e., user stories and acceptance criteria) of the target software that allows a human player to play a Solitaire game. These requirements will be fully implemented by the end of sprint 3. The minimum features include **choosing the board size and type, starting a new game, making a move, determining if a game is over**. The following is a sample GUI layout.

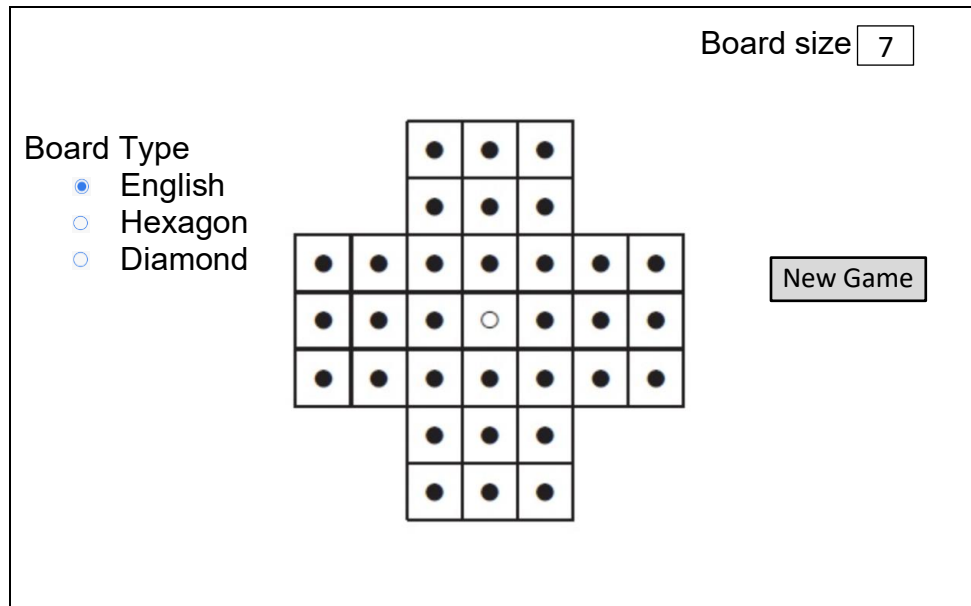


Figure 1. Sample GUI layout of the first working program by the end of Sprint 3

Use the following tables to document your user stories and acceptance criteria.

You are required to use the free ChatGPT or another LLM to complete two user stories and two of their respective acceptance criteria. You must ensure that the generated user stories are correct and fix them if not. At the end of the submission, provide screenshots of your LLM prompts and answers, along with the description of errors the LLM made that you had to correct. Points will be deducted if no screenshots/description are provided.

For the other user stories and acceptance criteria, you are not required to use LLM.

I. User Stories (3 points)

- **User Story Template:** As a <role>, I want <goal> [so that <benefit>]

ID	User Story Name	User Story Description	Priority	Estimated effort (hours)
1	Choose a board size	As a <player>, I want <to pick the board size> [so that <i can pick how long i want to play for>]	1/5	3
2	Choose the board type	As a <player>, I want <choose the game board style> [so that <i can choose my favorite or difficulty >]	3/5	4

ID	User Story Name	User Story Description	Priority	Estimated effort (hours)
3AI	Start a new game of the chosen board size and type	As a player, I want to start a new game by choosing a board size and board type so that I can play the version of marble solitaire I want.	5/5	6
4AI	Make a move in a game	As a player, I want to move a marble by jumping it over an adjacent marble into an empty hole so that I can progress toward clearing the board.	5/5	8
5	A game is over	As a <player>, I want <to know when the game is done> [so that <i can play again and know how good i did >]	4/5	5

II. Acceptance Criteria (AC) (12 points): Add/delete rows as needed.

User Story ID and Name	AC ID	Description of Acceptance Criterion	Status (completed, toDo, inProgress)
1. Choose a board size	AC 1.1	Bad board size. Given a board size that's too small or too big, when a user enters a board size outside the usable parameter, then clear the text box and give an error message of what they did wrong and the acceptable parameter so they can try again.	toDo
	AC 1.2	Good board size. Given an acceptable parameter for board size, when the user enters a board size, then say it's good and send the board size into the program to scale the board to the size selected.	toDo
	AC 1.3	Non-numeric entry. Given the user types letters, symbols, decimals, or negative numbers, when they submit the size, then clear the text box and show an error message saying a whole number between [min] and [max] is required.	toDo
	AC 1.4	Empty entry. Given the size field is blank, when the user submits, then show an error message saying a board size is required and do not change the current size.	toDo
	AC 1.5	Boundary values. Given the user enters exactly [min] or exactly [max], when they submit the size, then the size is accepted as valid.	toDo
	AC 1.6	Default size. Given the user has not chosen a size, when the game screen loads, then a default size is preselected and displayed so a game can still be started.	toDo
	AC 1.7	Size persists. Given a valid size was entered, when the user changes the board type or returns to the setup screen, then the chosen size is still displayed and remains selected.	toDo
	AC 1.8	Size change during a game. Given a game is in progress, when the user tries to change the board size, then the current game is not affected until a new game is started (see AC 3.6 for the confirmation prompt).	toDo
2. Choose the board type	AC 2.1	Select a board type. Given the list of available board types (e.g., English, European), when the user selects one, then that type is highlighted as the current selection and a preview or description of the layout is shown.	toDo

User Story ID and Name	AC ID	Description of Acceptance Criterion	Status (completed, toDo, inProgress)
	AC 2.2	Only one type at a time. Given a board type is already selected, when the user selects a different type, then the new type becomes selected and the previous one is deselected.	toDo
	AC 2.3	Default type. Given the user has not chosen a type, when the setup screen loads, then no type is selected (or a default is preselected, depending on your design decision) and the behavior matches AC 3.3.	toDo
	AC 2.4	Type and size compatibility. Given a board size and type are both selected, when the combination is not supported (e.g., a size that does not fit the type's layout), then show an error message explaining the conflict and the valid sizes for that type.	toDo
	AC 2.5	Type persists. Given a board type was selected, when the user changes the board size, then the selected type remains selected.	toDo
	AC 2.6	Type change during a game. Given a game is in progress, when the user tries to change the board type, then the current game is not affected until a new game is started (see AC 3.6 for the confirmation prompt).	toDo
	AC 3.1	Bad board size. Given a board size that is too small or too big, when the player enters a size outside the usable parameters ([min]–[max]), then clear the text box and show an error message stating what was wrong and the acceptable range, so they can try again.	toDo
3. Start a new game of the chosen board size and type AI	AC 3.2	Non-numeric board size. Given the player types letters, symbols, or leaves the field blank, when they try to start the game, then clear the text box and show an error message saying a whole number between [min] and [max] is required.	toDo
	AC 3.3	No board type selected. Given no board type is chosen, when the player clicks Start, then the game does not start and a message prompts them to select a board type.	toDo
	AC 3.4	Valid new game. Given a valid board size and board type, when the player clicks Start, then the board is displayed in the correct layout for that type, every hole is filled with a marble except the starting empty hole (center by default), the marble count is shown, and the move counter is set to 0.	toDo
	AC 3.5	Board matches chosen type. Given the player selects a specific board type (e.g., English or European), when the game starts, then the hole pattern and total marble count match that type's definition.	toDo
	AC 3.6	Game already in progress. Given a game is in progress, when the player starts a new game, then show a confirmation prompt. If they confirm, the current game is	toDo

User Story ID and Name	AC ID	Description of Acceptance Criterion	Status (completed, toDo, inProgress)
		discarded and the new one starts. If they cancel, the current game continues unchanged.	
4. Make a move in a game AI	AC 4.1	Valid move. Given a marble is selected and an empty hole is exactly two spaces away horizontally or vertically with a marble between them, when the player selects that empty hole, then the selected marble moves there, the jumped marble is removed, the marble count decreases by 1, and the move counter increases by 1.	toDo
	AC 4.2	Destination not empty. Given a marble is selected, when the player selects a hole that already contains a marble, then the move is rejected, the board is unchanged, and an error message says the destination must be empty.	toDo
	AC 4.3	No marble to jump. Given a marble is selected and the hole between it and the destination is empty, when the player selects that destination, then the move is rejected, the board is unchanged, and an error message says a marble must be jumped.	toDo
	AC 4.4	Illegal direction or distance. Given a marble is selected, when the player selects a destination that is diagonal or not exactly two spaces away, then the move is rejected, the board is unchanged, and an error message explains that moves must jump two spaces horizontally or vertically.	toDo
	AC 4.5	Invalid selection. Given the player clicks an empty hole or a spot outside the board as the first selection, when the click registers, then nothing is selected and a message tells them to select a marble.	toDo
	AC 4.6	Deselect marble. Given a marble is selected, when the player clicks the same marble again, then it is deselected and the board is unchanged.	toDo
5. A game is over	AC 5.1	Game not over. Given at least one legal move remains, when a move is completed, then the game continues, no game-over message is shown, and the board stays interactive.	toDo
	AC 5.2	Win (one marble left). Given a move leaves exactly one marble on the board, when the move is completed, then the game ends and a win message is shown with the final marble count of 1 and the total number of moves.	toDo
	AC 5.3	Perfect win (last marble in center). Given exactly one marble remains and it is in the starting center hole, when the game ends, then show a special "perfect solution" message in addition to the standard win message.	toDo
	AC 5.4	Loss (no moves left). Given more than one marble remains and no legal moves exist, when the last move is completed, then the game ends and a message is shown with the final marble count and a note that no moves remain.	toDo

User Story ID and Name	AC ID	Description of Acceptance Criterion	Status (completed, toDo, inProgress)
	AC 5.5	Performance rating. Given the game has ended, when the results are shown, then a rating based on the remaining marbles (e.g., 1 = Genius, 2–3 = Good, 4+ = Keep Trying) is displayed so the player knows how well they did.	toDo
	AC 5.6	Board locked after game over. Given the game has ended, when the player clicks a marble or hole on the board, then nothing happens and the board is unchanged.	toDo
	AC 5.7	Play again. Given the game-over screen is shown, when the player clicks Play Again, then a new game starts with the same board size and type, the marble count is reset, and the move counter is set to 0.	toDo
	AC 5.8	Return to setup. Given the game-over screen is shown, when the player clicks New Setup (or Back), then the setup screen is shown so they can pick a different board size and type.	toDo
	AC 5.9	Final stats displayed. Given the game has ended, when the game-over message is shown, then the final marble count, total moves made, and result (win or loss) are all displayed.	toDo

project is a marble solitaire game

User Story Template: ID:# | User Story Name | User Story Description: As a <role>, I want <goal> [so that <benefit>] | Priority #/5 | Estimated time to complete in hours : #

acceptance criteria Template: User Story ID and Name | Acceptance Criteria ID : User Story ID.# | Description of Acceptance Criterion : {the following is an example}

(AC 1.1 <bad board size>

Given a board size thats to small or big

When a user enters a board size outside of usable peramiter

Then clear text box and give an error message of what they did wrong to try again and the acceptable peramiter) | Status : (completed, toDo, inPprogress)

make a user story and however many acceptance Criteria needed for both of the following scenario

Start a new game of the chosen board size and type

Make a move in a game

Show less

28 minutes ago   

Drafting user stories for a marble solitaire game.

Marble Solitaire: User Stories and Acceptance Criteria

I assumed board type means layouts like English (33-hole) or European (37-hole), and I left the size limits as placeholders. Swap in your real values.

User Stories

ID	User Story Name	User Story Description	Priority	Est. Time (hrs)
1	Start New Game	As a player, I want to start a new game by choosing a board size and board type so that I can play the version of marble solitaire I want.	5/5	6
2	Make a Move	As a player, I want to n  a marble by jumping it over an adjacent marble into an	5/5	8

2	Make a Move	As a player, I want to move a marble by jumping it over an adjacent marble into an empty hole so that I can progress toward clearing the board.	5/5	8
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Acceptance Criteria

User Story 1: Start New Game

AC ID	Description	Status
AC 1.1	Bad board size. Given a board size that is too small or too big, when the player enters a size outside the usable parameters ([min]–[max]), then clear the text box and show an error message stating what was wrong and the acceptable range, so they can try again.	ToDo
AC 1.2	Non-numeric board size. Given the player types letters, symbols, or leaves the field blank, when they try to start the game, then clear the text box and show an error message saying a whole number between [min] and [max] is required.	ToDo
AC 1.3	No board type selected. Given no board type is chosen, when the player clicks Start, then the game does not start and a message prompts them to select a board type.	ToDo
AC 1.4	Valid new game. Given a valid board size and board type, when the player clicks Start, then the board is displayed in the correct layout for that type, every hole is filled with a marble except the starting empty hole (center by default), the marble count is shown, and the move counter is set to 0.	ToDo
AC 1.5	Board matches chosen type. Given the player selects a specific board type (e.g., English or European), when the game starts, then the hole pattern and total marble count match that type's definition.	ToDo
AC 1.6	Game already in progress. Given a game is in progress, when the player starts a new game, then show a confirmation prompt. If they confirm, the current game is discarded and the new one starts. If they cancel, the current game continues unchanged.	ToDo

User Story 2: Make a Move

AC ID	Description	Status
AC 2.1	Valid move. Given a marble is selected and an empty hole is exactly two spaces away horizontally or vertically with a marble between them, when the player selects that empty hole, then the selected marble moves there, the jumped marble is removed, the marble count decreases by 1, and the move counter increases by 1.	toDo
AC 2.2	Destination not empty. Given a marble is selected, when the player selects a hole that already contains a marble, then the move is rejected, the board is unchanged, and an error message says the destination must be empty.	toDo
AC 2.3	No marble to jump. Given a marble is selected and the hole between it and the destination is empty, when the player selects that destination, then the move is rejected, the board is unchanged, and an error message says a marble must be jumped.	toDo
AC 2.4	Illegal direction or distance. Given a marble is selected, when the player selects a destination that is diagonal or not exactly two spaces away, then the move is rejected, the board is unchanged, and an error message explains that moves must jump two spaces horizontally or vertically.	toDo
AC 2.5	Invalid selection. Given the player clicks an empty hole or a spot outside the board as the first selection, when the click registers, then nothing is selected and a message tells them to select a marble.	toDo
AC 2.6	Deselect marble. Given a marble is selected, when the player clicks the same marble again, then it is deselected and the board is unchanged.	toDo
AC 2.7	Game over detection. Given a move has just been completed, when no further legal moves remain, then the game ends and shows a message with the final marble count. One remaining marble is a win, and more than one is a loss.	toDo



If you'd like, I can add more stories (undo move, reset game, hints) or adjust the

these were good because i made sure to thoroughly set a templet and guidance criteria of what i wanted. It already knows user stories and acceptance criteria and many diffrent versions so i just made sure it knew how i wanted it presented to me and what for. 2.7 overlaps a bit with 5 so ill likely delete it. After word i used the same chat so the rules a set out stayed and gave it the user stories and acceptance criteria i had already made for the other 3 and let it add to them and re word the ones i made so they all sounded and look like they follow the same format.

User Story ID and Name
AC ID Description of Acceptance Criterion Status (completed, toDo, inProgress)
a) 1. Choose a board size
1 AC 1.1 <bad board size...

PASTED

add more Acceptance criteria to user stories 1, 2, 5

ID

User Story Name

User Story Description

Priority

Estimated effort (hours)

1 Choose a board size

Show more

Additional Acceptance Criteria: User Stories 1, 2, and 5

I kept your existing ACs (1.1, 1.2, 2.1, 5.1) and continued the numbering. Story 1 uses [min]-[max] placeholders for the size limits, so fill those in with your real values. Where your placeholder rows (2.1 and 5.1) were still blank, I filled them in.

User Story 1: Choose a Board Size

AC ID	Description	Status
AC 1.1	Bad board size. Given a board size that's too small or too big, when a user enters a board size outside the usable parameter, then clear the text box and give an error message of what they did wrong and the acceptable parameter so they can try again.	toDo
AC 1.2	Good board size. Given an acceptable parameter for board size, when the user enters a board size, then say it's good and send the board size into the program to scale the board to the size selected.	toDo
AC 1.3	Non-numeric entry. Given the user types letters, symbols, decimals, or negative numbers, when they submit the size, then clear the text box and show an error message saying a whole	toDo

not change the current size.

AC 1.5	Boundary values. Given the user enters exactly [min] or exactly [max], when they submit the size, then the size is accepted as valid.	ToDo
AC 1.6	Default size. Given the user has not chosen a size, when the game screen loads, then a default size is preselected and displayed so a game can still be started.	ToDo
AC 1.7	Size persists. Given a valid size was entered, when the user changes the board type or returns to the setup screen, then the chosen size is still displayed and remains selected.	ToDo
AC 1.8	Size change during a game. Given a game is in progress, when the user tries to change the board size, then the current game is not affected until a new game is started (see AC 3.6 for the confirmation prompt).	ToDo

User Story 2: Choose the Board Type

AC ID	Description	Status
AC 2.1	Select a board type. Given the list of available board types (e.g., English, European), when the user selects one, then that type is highlighted as the current selection and a preview or description of the layout is shown.	ToDo
AC 2.2	Only one type at a time. Given a board type is already selected, when the user selects a different type, then the new type becomes selected and the previous one is deselected.	ToDo
AC 2.3	Default type. Given the user has not chosen a type, when the setup screen loads, then no type is selected (or a default is preselected, depending on your design decision) and the behavior matches AC 3.3.	ToDo
AC 2.4	Type and size compatibility. Given a board size and type are both selected, when the combination is not supported (e.g., a size that does not fit the type's layout), then show an error message explaining the conflict and the valid sizes for that type.	ToDo

does not fit the type's layout), then show an error message explaining the conflict and the valid sizes for that type.

AC 2.5	Type persists. Given a board type was selected, when the user changes the board size, then the selected type remains selected.	toDo
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AC 2.6	Type change during a game. Given a game is in progress, when the user tries to change the board type, then the current game is not affected until a new game is started (see AC 3.6 for the confirmation prompt).	toDo
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User Story 5: A Game Is Over

AC ID	Description	Status
AC 5.1	Game not over. Given at least one legal move remains, when a move is completed, then the game continues, no game-over message is shown, and the board stays interactive.	toDo
AC 5.2	Win (one marble left). Given a move leaves exactly one marble on the board, when the move is completed, then the game ends and a win message is shown with the final marble count of 1 and the total number of moves.	toDo
AC 5.3	Perfect win (last marble in center). Given exactly one marble remains and it is in the starting center hole, when the game ends, then show a special "perfect solution" message in addition to the standard win message.	toDo
AC 5.4	Loss (no moves left). Given more than one marble remains and no legal moves exist, when the last move is completed, then the game ends and a message is shown with the final marble count and a note that no moves remain.	toDo
AC 5.5	Performance rating. Given the game has ended, when the results are shown, then a rating based on the remaining marbles (e.g., 1 = Genius, 2-3 = Good, 4+ = Keep Trying) is displayed so the player knows how well they did.	toDo
AC 5.6	Board locked after game over. Given the game has ended, when the player clicks a marble or hole on the board, then nothing	toDo

ends and a message is shown with the final marble count and a note that no moves remain.

AC 5.5	Performance rating. Given the game has ended, when the results are shown, then a rating based on the remaining marbles (e.g., 1 = Genius, 2-3 = Good, 4+ = Keep Trying) is displayed so the player knows how well they did.	ToDo
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AC 5.6	Board locked after game over. Given the game has ended, when the player clicks a marble or hole on the board, then nothing happens and the board is unchanged.	ToDo
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AC 5.7	Play again. Given the game-over screen is shown, when the player clicks Play Again, then a new game starts with the same board size and type, the marble count is reset, and the move counter is set to 0.	ToDo
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AC 5.8	Return to setup. Given the game-over screen is shown, when the player clicks New Setup (or Back), then the setup screen is shown so they can pick a different board size and type.	ToDo
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AC 5.9	Final stats displayed. Given the game has ended, when the game-over message is shown, then the final marble count, total moves made, and result (win or loss) are all displayed.	ToDo
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A couple of notes:

A couple of notes:

- **Story 4 overlap:** AC 4.7 (game over detection) and AC 5.1 to 5.4 cover related ground. You could keep 4.7 as the *detection* trigger and let Story 5 own everything the player sees afterward.
- **AC 2.3:** This depends on a design choice. Decide whether a default type is preselected, then trim the wording accordingly.

i simply deleted 4.7 like i mentioned before. Other than that they all seem good to me after reading through them. It came up with a few things i didn't even think of like 2.6. i dont plan on there being a way to change the type in the middle of the game so it never accured to me that i should put that down considering this is supposed to be before the game is made. Quite a few of them were that way actually. They felt like common since of course you need to make it in a way that these cant be a problem but this isn't for me. If i make it its supposed to be for the team and help them make sure everything works properly so i would have left way to much stuff out. It really helped me get a better understanding of the perpose of the user story and AC.